# Machine Learning: glossary

Every word in the course, with the session where it first appears. This list is generated from the "Words for today" cards, so a word cannot be taught without reaching this handout.

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| accuracy | The share of predictions that are right. | Marks scored out of the total. | ML16 |
| actual and predicted | Rows show what really happened; columns show what the model said. | The real result against the guess. | ML17 |
| AI use log | A record of any assistance, advice taken or rejected, and tests run. | A signed note of who helped. | ML21 |
| argument | The actual value you give the function when you use it. | What you write in the blank. | ML06 |
| assert | Stops with an error if a statement is false. | A teacher checking that an answer is right. | ML06 |
| audience | The people listening and what they already know. | The parents in the room. | ML23 |
| axis label | Words that name what an axis measures, with its unit. | The heading on a column in a register. | ML11 |
| backup | A screenshot or saved file you can show if the laptop fails. | A spare key. | ML23 |
| bar chart | Bars compare a number across categories. | Stacks of coins, one stack per item. | ML11 |
| baseline | A simple guess a model must beat, such as the training mean. | Guessing that every home costs the average rent. | ML13 |
| baseline error | The average miss of the plain guess on validation. | How wrong the average guess is. | ML20 |
| baseline model | A model that guesses the training mean every time. | The average guess. | ML15 |
| Boolean | A value that is either True or False. | A light switch that is on or off. | ML04 |
| box plot | A box shows the middle half of the values, with a line at the median. | A summary of a class's marks in one picture. | ML11 |
| candidate model | A model you are considering using. | A player trying out for the team. | ML21 |
| check | A test with a known right answer. | Weighing a bag of sugar on a trusted scale. | ML06 |
| checkpoint | A piece of work that shows what you can do alone so far. | A mid-term test you can rerun. | ML12 |
| claim | A statement that needs a visible result. | I say the model predicts fares to within about UGX 3000. | ML23 |
| class label | The name of a category; its code number is only an identifier. | The species name on a seed packet. | ML16 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| classification | Predicting a category instead of a number. | Deciding whether a fruit is a mango or an orange. | ML16 |
| cleaning log | A list of every change made and why. | The marking notes beside a corrected exercise. | ML10 |
| code cell | A box where you type Python and press Shift+Enter to run it. | A calculator that sits inside your page. | ML01 |
| coding assistant | A tool that suggests code or fixes; its advice must be checked. | A friend who sounds sure and is sometimes wrong. | ML21 |
| comment | A line starting with # that Python ignores and people read. | A note written in the margin. | ML03 |
| comparison | A question about two values that answers True or False. | Asking whether your money is at least the ticket price. | ML04 |
| completion record | A note of the track finished and the evidence assessed. | A school report. | ML24 |
| confusion matrix | A table that counts actual categories against predicted categories. | A mark sheet of right and wrong answers by question type. | ML17 |
| convert | Change a value from one type to another, such as text to a number. | Changing shillings into dollars before adding. | ML03 |
| CSV file | A plain text file where each line is a row and commas split the columns. | A register copied as text. | ML07 |
| data dictionary | A list of each column with its meaning and unit. | The key at the bottom of a map. | ML12 |
| DataFrame | The pandas name for a table inside Python. | A spreadsheet that lives in the notebook. | ML07 |
| decision tree | A chain of yes or no questions about the features that ends in a category. | A flow chart that sorts the flowers. | ML16 |
| demonstration | Showing your own work running or explaining its steps. | Showing the dish you cooked. | ML23 |
| depth | How many questions in a row the tree may ask. | How many forks a path may take. | ML16 |
| dictionary | Pairs of a name and a value inside curly brackets. | A phone book of names and numbers. | ML05 |
| disclosure | Saying where help or AI was used. | Thanking those who helped. | ML23 |
| duplicate | A row repeated exactly. It may be an accident or a real second sale. | The same receipt copied twice. | ML10 |
| elif | Asks a second question when the first was False. | If it is raining take an umbrella, otherwise if it is cloudy take a hat. | ML04 |
| else | Runs when the if question was False. | Otherwise, leave the umbrella. | ML04 |
| error | The gap between a prediction and the real value. | How far off your estimate was. | ML14 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| error message | Python telling you what it could not do and where. | A teacher marking the exact line you got wrong. | ML03 |
| fair comparison | Same rows, same measure, same split for every model. | Every runner on the same track. | ML21 |
| feature | An input known before you predict. | The clues in a puzzle. | ML13 |
| feedback | A strength and a next step from the instructor. | A teacher's comment. | ML24 |
| file | A named container for your work on the laptop. | A paper file in a folder at school. | ML01 |
| filter | Keep only the rows where a comparison is True. | Picking out the sales of 5 units or more. | ML08 |
| final test | The last check, run once after every choice is made. | The real exam. | ML17 |
| finding | A claim a chart supports, with a number. | Book earns the most money: UGX 297000. | ML11 |
| fit | Let the computer find the best slope and intercept from examples. | Practising until the throws land close. | ML14 |
| float | A number with a decimal part. | A price of 2500.5 shillings. | ML02 |
| frozen choice | A decision fixed before the final test is run. | Handing in your answer sheet before seeing the memo. | ML18 |
| frozen model | A model whose inputs and settings are fixed before the final test. | An answer sheet handed in. | ML22 |
| function | A named set of instructions you can use again and again. | A recipe card you reuse. | ML06 |
| group | Gather rows with the same label before calculating. | Sorting the sales into three piles by item. | ML08 |
| handover | Everything another person needs: data origin, code, instructions and limits. | Passing on the keys with directions. | ML22 |
| header | The name written at the top of each column. | The title above a column in an exercise book. | ML07 |
| held-out records | The rows kept back and never used to fit. | The exam paper kept in a locked cupboard. | ML15 |
| histogram | Bars show how many values fall in each range. | Sorting pupils into groups by height. | ML11 |
| if | Runs the indented lines below it only when its question is True. | If it rains, take an umbrella. | ML04 |
| indentation | Four spaces at the start of a line that show which instructions belong together. | Writing a sub-point under a heading. | ML04 |
| index | The position of an item in a list, counting from 0. | The house number of a plot. | ML05 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| index=False | Tells to_csv not to write the extra row-number column. | Not copying the page numbers. | ML10 |
| input | A measurement known before you predict the target. | The clues you are given. | ML12 |
| inputs | The columns known before the prediction. | The clues you are given. | ML19 |
| integer | A whole number with no decimal part. | Counting 3 books. | ML02 |
| intercept | The prediction when the input is zero. | The starting fee before any distance is added. | ML14 |
| item | One value taken from a list during a loop. | The pupil currently getting a book. | ML05 |
| key | The name used to look up a value in a dictionary. | The name you look up in a phone book. | ML05 |
| leakage | Using information that would not be known at prediction time. | Seeing the exam answers before the exam. | ML13 |
| leakage column | A column only known after the answer, so it must not be an input. | A tip that is only known after the ride. | ML19 |
| library and import | A library is code other people wrote. import loads it into your notebook. | Borrowing a toolbox from the workshop. | ML07 |
| limit | A fair statement of what the result cannot show. | The edge of what you know. | ML22 |
| limitation | What the data or method cannot tell you. | The edge of the map. | ML12 |
| limits | What the data and the model cannot tell us. | The edges of the map. | ML19 |
| list | Several values kept in order inside square brackets. | A queue of customers. | ML05 |
| locked test | A test set no choice has ever been based on. | An exam paper no student has seen. | ML17 |
| loop | A for statement that repeats indented lines for every item. | Giving each pupil in a row a book. | ML05 |
| majority baseline | A model that always names the most common category. | Always answering the most popular option. | ML16 |
| mean | The total shared out equally: add the values and divide by how many. | Sharing a bag of mandazi equally. | ML09 |
| mean absolute error | The average size of the misses, in the target's own unit. | The average distance of your darts from the bullseye. | ML15 |
| median | The middle value once the values are in order. | The pupil standing in the middle of a line sorted by height. | ML09 |
| missing value | A cell with no recorded value. Its true value is unknown. | A blank in the price column. | ML10 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| new column | A column calculated from other columns, added to the table. | Multiplying units by price on every line. | ML08 |
| next step | One learning goal for after the camp. | The next chapter you plan to read. | ML24 |
| notebook | A page made of text cells that explain and code cells that run. | An exercise book where the answers can run themselves. | ML01 |
| operator | A symbol that tells Python which calculation to do. | The plus and minus keys on a calculator. | ML03 |
| outlier | A value far from the rest. | One huge order in a week of small sales. | ML09 |
| output | What Python shows under a cell after it runs. | The answer a calculator shows after you press equals. | ML01 |
| overfitting | Learning the quirks of the training examples so well that new examples are predicted worse. | Memorising last year's exam instead of learning the subject. | ML17 |
| parameter | The name inside the function for the value it receives. | The blank on a form. | ML06 |
| parentheses | Brackets that make Python calculate the inside first. | Doing the part in brackets first in maths. | ML03 |
| percentile | The value below which a given share of the data falls. | The mark that 75 out of 100 pupils did not beat. | ML09 |
| portfolio | The folder of your own project files and feedback. | Your best exercise books. | ML24 |
| predict | Use the fitted line on a new input. | Reading the answer off the line. | ML14 |
| prediction | A number or label a computer guesses for something it has not seen yet. | A fundi estimating a job before measuring it. | ML01 |
| print | A ready-made instruction that shows a value on the screen. | Reading a number out loud. | ML01 |
| project plan | A short page that fixes the question, data, target, inputs, baseline, test and limits before any building. | A drawing of the house before laying bricks. | ML19 |
| proportion | A part divided by the whole; times 100 it becomes a percentage. | 12 pupils out of 20 passed. | ML09 |
| random_state | A number that makes the shuffle repeatable. | Giving everyone the same shuffled deck. | ML15 |
| range | The largest value minus the smallest. | The gap between the tallest and the shortest. | ML09 |
| raw data | The file as you received it. Keep it unchanged. | The original receipt book. | ML10 |
| regression | Predicting a number. | Estimating a price. | ML14 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| reproducible | Someone can rerun your notebook from the top and get the same results. | A recipe anyone can follow. | ML12 |
| restart and run all | Restart the kernel and run every cell from the top. | Starting the game from the first level. | ML22 |
| return | Sends a result back so the program can keep it. | Handing over the finished item. | ML06 |
| revenue | Money received from sales: units times price. It is not profit. | The takings before paying for stock. | ML08 |
| row and column | A row is one record across the table. A column is one kind of fact down the table. | One line of a register and one heading of the register. | ML07 |
| scatter plot | Each row is a dot; its place shows two values. | Dots on a map of distance and fare. | ML20 |
| select a column | Name the column in square brackets to look only at it. | Reading only the price line of the register. | ML08 |
| shape | The number of rows and columns, written as a pair. | Counting how many lines and headings a register has. | ML07 |
| showcase | A short event where you present your project to parents and guardians. | A school open day. | ML24 |
| slope | How much the prediction changes when the input rises by one unit. | Extra rent for every extra square metre. | ML14 |
| split sizes | How many rows go to training, validation and test. | How the class is divided for the three jobs. | ML18 |
| stratify | Keep the share of each category the same in every part. | Making sure every group has some of each village. | ML18 |
| string | Text inside quotation marks. | A name written on a label. | ML02 |
| sum | All the values added together. | Adding up the day's takings. | ML08 |
| target | The value you want to predict, such as monthly rent. | The number you are guessing. | ML12 |
| test data | Examples held back to check the model on things it has not seen. | The exam paper. | ML13 |
| test_size | The share of rows kept back for testing. | One quarter of the class sits the check. | ML15 |
| three-way split | Training, validation and final test parts. | Practice, mock and real exam. | ML20 |
| title | One line saying what the chart shows. | The heading of a notice. | ML11 |
| to_csv | Saves a table into a CSV file. | Copying the register to a new page. | ML10 |
| traceback | The list of lines Python followed before the error. | A trail of footprints that ends at the problem. | ML03 |

| Word | What it means | Think of it like this | Session |
|---|---|---|---|
| train and test split | Divide the rows into homes to learn from and homes to test on. | Practice questions and the real exam. | ML15 |
| training data | The examples the model learns from. | The worked examples in a textbook. | ML13 |
| training only | Fit anything only on the training part. | Learning only from the textbook. | ML20 |
| type | The kind of value: integer, float or string. | Knowing whether a box holds pens or juice. | ML02 |
| unaided | Done with no AI assistance, with assistant windows closed. | A closed-book exam. | ML24 |
| validation error | The average miss on validation rows. | The mock exam mark. | ML21 |
| validation records | Held-out rows used to choose between models. | The practice test used to decide how to revise. | ML17 |
| validation score | A score on held-out rows used to choose between models. | A mock exam. | ML18 |
| value | One piece of information, such as 2000 or Juice. | One entry in a receipt. | ML02 |
| variable | A name that points to a value. | A label on a box of stock. | ML02 |